

Getting Started with MongoDB



Objectives

After completing this lab you will be able to:

- Access the MongoDB server using the command-line interface
- Describe the process of listing and creating collections, which contain documents, and databases, which contain one or more collections
- Perform basic operations on a collection such as inserting, counting and listing documents

Exercise 1 - Start mongodb server

Open the MongoDB database page by clicking the button below:

[Open MongoDB Page in IDE](#)

On that page, click the **Create** button to create a MongoDB database.

[←](#) [→](#)

The screenshot shows a sidebar with a navigation menu (Databases, Big Data, Cloud, Embeddable AI, Other) and a main panel titled "MongoDB" with a status of "INACTIVE". It includes version information (3.6.3, 0.54.0, 2.0.2) and a "Create" button.

Exercise 2 - Connect to mongodb server

After creation has finished, click the **MongoDB CLI** button (below the **Create** button from the previous step) to connect to the database using the mongosh CLI.

You should now get connected to the mongodb database, and see an output as in the figure below.

```
theia@theiadoser-naimbenalaya:/home/project$ mongosh mongodb://root:1UT3QIGpN50921YqUSmPAw@172.21.78.21:27017
Current Mongosh log ID: 66f3ab566a7b59c27964032
Connecting to:
  mongodb://root:1UT3QIGpN50921YqUSmPAw@172.21.78.21:27017/?directConnection=true&appName=mongosh+2.3.1
Using MongoDB:
  3.6.3
Using Mongosh:
  2.3.1

For mongosh info see: https://www.mongodb.com/docs/mongosh-shell/

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (https://www.mongodb.com/legal/privacy-policy).
You can opt-out by running the disableTelemetry() command.

*****
The server generated these startup warnings when booting
2024-09-25T03:28:13.806+0000: ** WARNING: Using the XFS filesystem is strongly recommended with the WiredTiger
  storage engine
2024-09-25T03:28:13.806+0000: ** See http://docs.mongodb.org/manual/notes-on-filesystem
*****

text> 
```

The screenshot shows the same sidebar as Exercise 1, but the main panel now displays a "Summary" section with instructions on how to connect to the database using the MongoDB CLI or a new terminal. The "MongoDB CLI" button is highlighted.

Exercise 3 - Find the version of the server

On the mongo client run the below command.

```
1 db.version()
```

This will show the version of the mongodb server.

```
theia@theiadocker-naimbenalaya:/home/project$ mongosh mongodb://root:Mz18982Zdv13ulASL12S3DFf@172.21.94.229:27017
Current Mongosh Log ID: 686ea873677ceb71126b140a
Connecting to:      mongodb://<credentials>@172.21.94.229:27017/?directConnection=true&appName=mongosh+2.4.2
Using MongoDB:      4.4.29
Using Mongosh:      2.4.2
```

For mongosh info see: <https://www.mongodb.com/docs/mongodb-shell/>

To help improve our products, anonymous usage data is collected and sent to MongoDB periodically (<https://www.mongodb.com/legal/privacy-policy>).

You can opt-out by running the `disableTelemetry()` command.

```
-----
The server generated these startup warnings when booting
  2025-07-09T15:39:08.075+00:00: Using the XFS filesystem is strongly recommended with the WiredTiger storage engine. See http://dochub.mongodb.org/core/prodnotes-filesystem
-----
```

```
test> db.version()
4.4.29
test> █
```

Exercise 4 - List databases

On the mongo client run the below command.

```
1 show dbs
```

This will print a list of the databases present on the server.

```
test> show dbs
admin    100.00 KiB
config   12.00 KiB
local    72.00 KiB
test> █
```

Exercise 5 - Create database

On the mongo client run the below command.

```
1 use training
```

This will create a new database named training. If a database named training already exists, it will start using it.

Exercise 6 - Create collection

On the mongo client run the below command.

```
1 db.createCollection("mycollection")
```

This will create a collection name mycollection inside the training database.

Exercise 7 - List collections

On the mongo client run the below command.

```
1 show collections
```

This will print the list of collections in your current database.

Exercise 8 - Insert documents into a collection

On the mongo client run the below command.

```
1 db.mycollection.insert({"color":"white","example":"milk"})
```

The above command inserts the json document {"color":"white","example":"milk"} into the collection.

Let us insert one more document.

```
1 db.mycollection.insert({"color":"blue","example":"sky"})
```

The above command inserts the json document {"color":"blue","example":"sky"} into the collection.

Insert 3 more documents of your choice.

Exercise 9 - Count the number of documents in a collection

On the mongo client run the below command.

```
1 db.mycollection.countDocuments()
```

This command gives you the number of documents in the collection.

Exercise 10 - List all documents in a collection

On the mongo client run the below command.

```
1 db.mycollection.find()
```

This command lists all the documents in the collection `mycollection`

Notice that mongodb automatically adds an `'_id'` field to every document in order to uniquely identify the document.

```
training> db.mycollection.insert({"color":"white","example":"milk"})
DeprecationWarning: Collection.insert() is deprecated. Use insertOne, insertMany, or bulkWrite.
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('686ea952677ceb71126b140b') }
}
training> db.mycollection.insert({"color":"blue","example":"sky"})
{
  acknowledged: true,
  insertedIds: { '0': ObjectId('686ea957677ceb71126b140c') }
}
training> db.mycollection.countDocuments()
2
training> db.mycollection.find()
[
  {
    _id: ObjectId('686ea952677ceb71126b140b'),
    color: 'white',
    example: 'milk'
  },
  {
    _id: ObjectId('686ea957677ceb71126b140c'),
    color: 'blue',
    example: 'sky'
  }
]
training> █
```

Exercise 11 - Disconnect from mongodb server

On the mongo client run the below command.

```
1 exit
```

Practice exercises

1. Problem:

List databases.

▼ Click here for Hint

*Use the 'show' command with *dbs* option.*

▼ Click here for Solution

```
1 show dbs
```



2. Problem:

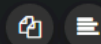
Create a database named mydatabase.

▼ Click here for Hint

Use the 'use' command with the database name.

▼ Click here for Solution

```
1 use mydatabase
```



3. Problem:

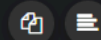
Create a collection named landmarks in the database mydatabase.

▼ Click here for Hint

Use the 'createCollection' command.

▼ Click here for Solution

```
1 db.createCollection("landmarks")
```



4. Problem:

List collections

▼ Click here for Hint

Use the 'show' command with collections_ option.

▼ Click here for Solution

```
1 show collections
```



5. Problem:

Insert details of five landmarks including name, city, and country. Example: Eiffel Tower, Paris, France.

▼ Click here for Hint

Use the 'db.collection.insert()' command with the correct options.

▼ Click here for Solution

```
1 db.landmarks.insert({"name":"Statue of Liberty","city":"New York","country":"USA"})
2 db.landmarks.insert({"name":"Big Ben","city":"London","country":"UK"})
3 db.landmarks.insert({"name":"Taj Mahal","city":"Agra","country":"India"})
4 db.landmarks.insert({"name":"Pyramids","country":"Egypt"})
5 db.landmarks.insert({"name":"Great Wall of China","country":"China"})
```

6. Problem:

Count the number of documents you have inserted.

▼ Click here for Hint

Use the 'count' command on your collections.

▼ Click here for Solution

```
1 db.landmarks.countDocuments()
```


7. Problem:

List the documents.

▼ Click here for Hint

Use the 'db.collection.find()' command.

▼ Click here for Solution

1

```
db.landmarks.find()
```



8. Problem:

Disconnect from the server.

▼ Click here for Hint

Use the 'exit' command.

▼ Click here for Solution

1

```
exit
```

